



Technical Appendix for: ***Catching Up and Moving Forward***

August 2022

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Introduction

As a nonprofit educational organization used by 1 in 4 elementary-school students and more than 1 million middle-school students nationwide, Zearn pursued this research to understand how teachers could most effectively respond to pandemic-related missed learning in a way that helps students catch-up and move forward in math. Every day, teachers make a multitude of decisions on how to best meet the needs of students in their classrooms. These challenges have been exacerbated by rising rates of missed learning, as evidenced by the large drops in achievement scores nationwide (NCES 2021). This analysis aimed to unearth a framework that teachers and school leaders could use to guide instructional decisions and support students' academic growth.

In spring 2021, Zearn and TNTP co-published [Accelerate, Don't Remediate](#), which detailed Zearn's analysis comparing two different approaches to address missed learning: learning acceleration vs. remediation. The analysis found that students who experienced learning acceleration (connecting unfinished learning into the context of new learning, integrating a few lessons from prior grades/units) repeatedly struggled less and learned more than students who started at the same level of initial repeated struggle but experienced remediation (broader, more general prior grade-level content covering many objectives or standards from prior grades/units before grade level) instead.

That previous analysis leveraged data from more than 6,000 third- through fifth-grade classrooms (serving 50K+ students) who, because of school closures, missed an entire section of critical math content during the 2019-2020 school year and had at least three years of historical, continuous Zearn platform usage to allow for analyzing trends over time. The report detailed the experiences of two groups during the fall of the 2020-21 school year: (1) those who experienced learning acceleration, and (2) those who experienced remediation, and found notable differences between the two groups: the different strategies led to a difference of 4-5 repeated struggle alerts¹ per lesson across a classroom of 20 students, and a 27% difference in grade-level progress as measured in December 2020.

Accelerate, Don't Remediate focused on a cohort of students during the fall of the 2020-21 school year. Now, with the benefit of two years of pandemic data, this report examines if a learning acceleration strategy still showed a relative advantage in reduced student repeated struggle and increased completion levels as compared to a remediation strategy. [Catching up and Moving Forward: Accelerating Math Learning for Every Student](#) discusses the results of this second study in looking at longer term impacts of the two strategies. This technical appendix provides further information about the sample, methods and results.

¹ Zearn's platform features digital math lessons designed to complement daily instruction. At the end of each lesson, a student demonstrates understanding through a daily digital, formative assessment called the [Tower of Power](#). If a student gets any question wrong, they receive in-the-moment support. If a student continues to struggle in the Tower of Power after multiple attempts, their teacher receives an alert in the [Tower Alerts Report](#), enabling them to provide differentiated support for that student. Throughout the paper, the phrase "repeated struggle" is used in reference to situations in which a Tower Alert was generated on the Zearn platform.

Research Questions

The primary research questions of this study were:

1. Which strategy, learning acceleration or remediation, appears to have greater efficacy in reducing students' repeated struggle on grade-level lessons? Do grade-level lesson completion rates vary by strategy?
2. Do learning acceleration and remediation show similar levels of efficacy in schools that primarily serve disadvantaged populations of students? Are students in these schools more-likely to experience remediation vs. learning acceleration?

Data and Sample Characteristics

For this analysis, researchers included anonymized usage data of millions of students on the Zearn platform during the 2020-2021 and 2021-2022 school years. Students qualified for inclusion in the data set if they had any instances of repeated struggle during the study period. Because the study's outcome of interest was repeated struggle rates following remediation vs. those rates following learning acceleration, only students who repeatedly struggled fit within the sampling frame. The composition of the sample, by subgroup, was within five percentage points of the overall Zearn user-base for every major demographic category.

For this analysis, interventions were categorized as “learning acceleration” or “remediation” based on the specific content teachers assigned in response to student repeated struggle. In the 2020-2021 academic year, Zearn introduced [Foundational Lesson Guidance](#), suggesting lessons for targeted support that helps students strengthen their understanding and get back to grade-level content as soon as possible. “Learning acceleration” is those situations in which either (a) students repeatedly struggled and were assigned these foundational lessons before returning to grade-level content, or (b) students repeatedly struggled and subsequently completed the digital lesson after receiving [built-in support](#) from the platform, and automatically progressed to the next grade-level lesson. “Remediation” is situations in which students repeatedly struggled and were assigned lessons that were below grade-level and isolated from grade-appropriate learning, before subsequently returning to grade-level content (see Figure 4 for more detail).

The vast majority of learning acceleration interventions in the data were automatically implemented by the Zearn platform, not by proactive teacher intervention. Although the phrase “teacher decisions” is used throughout, the learning acceleration strategy is primarily informed by data on an intervention that occurred when no foundational lessons were assigned by a student's instructor. The data set contained 4,935,636 instances of automated, built-in learning acceleration and 40,545 instances of teacher-initiated learning acceleration using Zearn's Foundational Lesson Guidance.

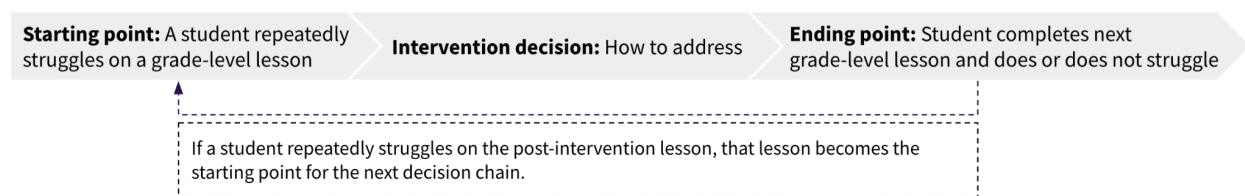
Therefore the reference to “teacher decisions” is not always reflective of an active process captured in the data, but rather of how a student spent their learning time after repeated struggle, dictated either by the software or by their teacher. However, the phrase is used because the aim of this analysis is to

inform strategies for students beyond their experiences on the platform, the preponderance of which must be implemented through action on the part of the teacher (see Descriptive Table 1 for a full breakdown of intervention frequency).

The analysis focused on the experiences of 626,653 students in grades 1-7, during the 2020-2021 and 2021-2022 school years. These students completed a mean of 46 lessons per user and received 5,279,131 teacher intervention decisions that took place after a student repeatedly struggled on grade-level content.

The unit of analysis for this study was a teacher decision that was operationalized by a chain of three events: 1) an on-grade lesson in which a student repeatedly struggled; 2) an intervention of learning acceleration or remediation; and 3) completion of the next on-grade lesson (see Figure 1 below). Because each decision chain starts with a lesson on which a student repeatedly struggled, the post-lesson on one chain could become the pre-lesson on the next chain if it was a lesson on which a student repeatedly struggled.

Figure 1. “Teacher decision” unit of analysis



Because repeated struggle on post-intervention lessons was the outcome of interest, instances in which a student repeatedly struggled and received an intervention, but then did not complete an additional grade-level lesson were excluded from the fixed effects analysis. Therefore, if the intervention itself discouraged students from continuing on the platform, that is not measured in this model. However, it would be challenging to disentangle whether usage stopped because of repeated struggle vs. many other possible explanations including the end of a school year, semester or other natural break in usage associated with the pacing of a class rather than the experience of an individual student.

In cases where students repeatedly struggled on their final grade-level lesson and then did not continue to complete any additional grade level lessons, remediation interventions were disproportionately represented. In the sample generally, remediation interventions comprised 6% of all interventions. However, among students who stopped using the Zearn platform following repeated struggle, remediation comprised 15% of interventions. Because of this, the fixed effects model likely underestimates the negative impact of remediation relative to learning acceleration (see Descriptive Tables 1 and 2).

While Zearn collects usage data on individual students, it does not collect demographic or socio-economic variables. Using MDR and publicly available school data, students were linked to school-level demographics. All demographic and economic subgroup analyses were conducted based

on the race/ethnicity or economic composition of users' schools. Therefore, subgroup analyses do not refer to students' race/ethnicity and economic position, rather to the experiences of students in schools with the indicated majority population based on race/ethnicity or income status. The sample spans all 50 states, and includes schools with a wide-range of economic and race/ethnicity composition (see Descriptive Table 3 for a full sample breakdown).

Question 1: Strategies, completion rates and the probability of repeated struggle

A quasi-experimental approach was used to isolate the impact of learning acceleration vs. remediation on student rates of repeated struggle, mediate confounding factors that arise when comparing across a population of students, and facilitate causal inference.

The first analysis for Question 1 examined all instances in which students repeatedly struggled while completing a grade-level lesson, how a teacher intervened to support the student's learning and whether or not they repeatedly struggled on the next grade-level lesson. Each instance was considered a "teacher decision" and, in total, there were 5,279,131 decisions during school years 2020-21 and 2021-22 (see Figure 1). This analysis investigated the probability of subsequent repeated struggle based on the intervention selected.

While the previous analysis, *Accelerate, Don't Remediate*, compared the impact of learning acceleration vs. remediation on similarly struggling students, this analysis utilized a logistic regression fixed effects model, which compares a student to themselves at multiple instances throughout the study period, rather than to other students in the population with whom they may have unaccounted for differences. In essence, fixed effects models use students as their own controls (Allison, 2009).

This analysis focused on the relative frequency of repeated struggle a student experienced after a learning acceleration strategy vs. after a remediation strategy, and averaged these within student differences across the population.

The standard fixed effects logit model is defined as the logistic probability of binary variable y_{it} such that:²

$$E[y_{it}|x_{it}, \alpha_i] = \Pr[y_{it} = 1|x_{it}, \alpha_i] = \frac{\exp(\alpha_i + \beta x_{it})}{1 + \exp(\alpha_i + \beta x_{it})}. \quad (\text{Figure 2})$$

Where $i = 1, \dots, N$ individuals for $t = 1, \dots, T$ time periods. In standard logit models, one unit change is not meaningful in the same way as in linear models because the partial effects are nonlinear combinations of all regressors and regression coefficients of the model. As a result, the coefficients themselves have little value in estimating the change in probability associated with a unit change in an

² Equation borrowed from Silva & Kemp's (2016) supplementary material found here: http://repec.org/usug2016/santos_uksug16.pdf

independent variable (Cornelißen, 2009). In contrast, researchers are generally interested in the change in probability associated with a one-unit change in explanatory variables, i.e., the marginal effects or semi-elasticities.

In order to generate conceptually meaningful coefficients, analysis was conducted using aextlogit, a wrapper for STATA's xtlogit command. This program estimates the fixed effects logit and reports estimates of the average (semi-) elasticities of:

$$\Pr [y_{it} = 1 | x_{it}, \alpha_i = 0]$$

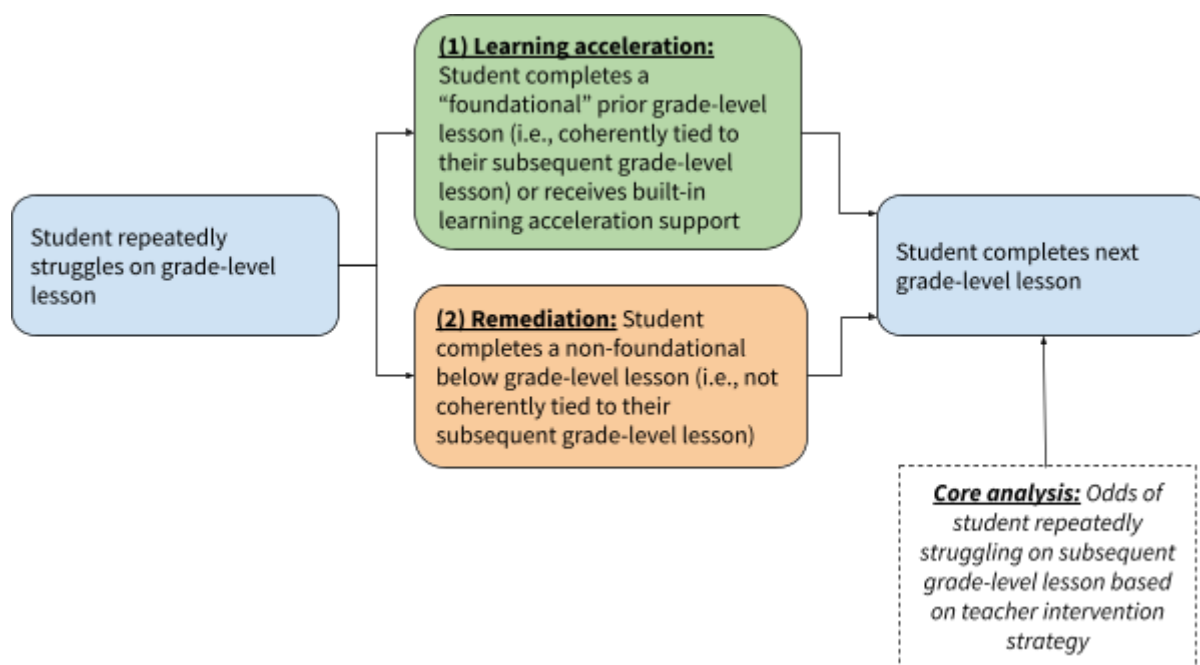
(Figure 3)

This command uses Kitazawa's method to compute semi-elasticities (with corresponding standard errors and t-statistics) for regressors in fixed effects logit models (Kitazawa, 2012; Kemp, 2016).

The aextlogit wrapper effectively estimates the average change in probability of subsequent repeated struggle based on a one-unit change in causal variables (in this case, the difference between remediation and learning acceleration). This method is particularly appropriate for consistently estimating the fixed effects logit model with a small number of time periods and large cross-sectional size, both characteristics of this data set, as fixed effects panel models provide the most consistent estimates as T approaches infinity.

The logistic fixed effects model was used to calculate the probability of a student repeatedly struggling based on their performance on the prior on-grade lesson and the strategy their teacher selected (see Figure 4).

Figure 4. Structure of Question 1's fixed effects logistic regression analysis

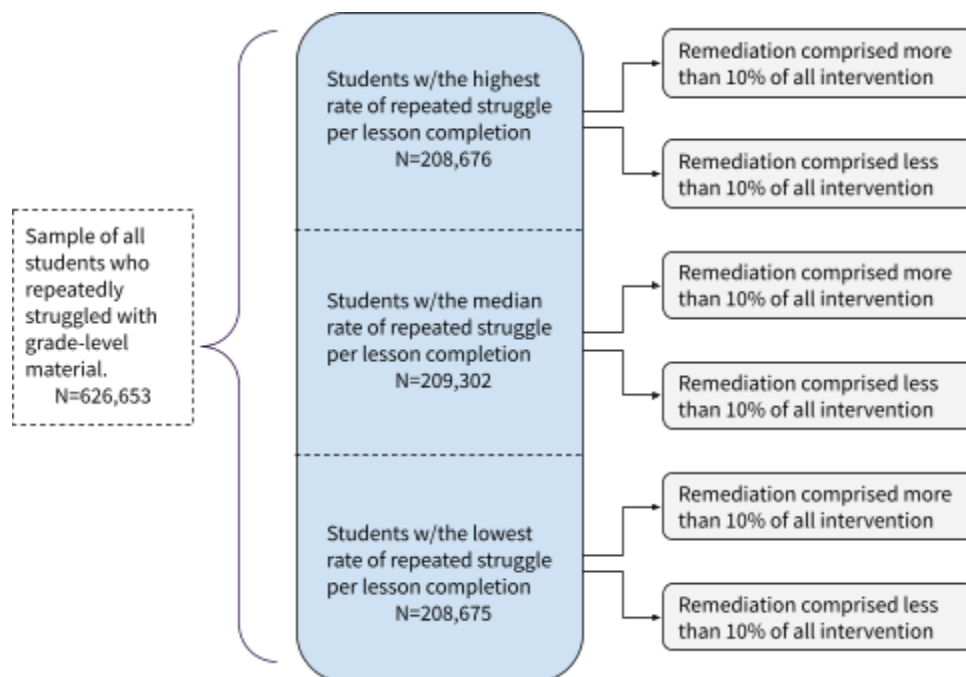


Findings showed that among all students in the sample, remediation interventions to students' repeated struggle were associated with a 44% probability of repeated struggle in subsequent lessons, while learning acceleration interventions resulted in a 36% probability of repeated struggle on the subsequent grade-level lesson; a 17% decrease in the probability of continued repeated struggle. This provided promising evidence that, as shown in *Accelerate, Don't Remediate*, learning acceleration remains a more effective intervention for repeated struggles with grade-level material (see Results Table 1 for more detail).

The second analysis for Question 1, investigated the relationship between frequency of teacher strategy and total lesson completion among students with similar levels of repeated struggle. Students who receive frequent intervention in the form of remediation or learning acceleration both spend time on below grade-level material to recover missed learning. If effective, this time on below grade-level material should reduce students' future repeated struggle and facilitate completion of grade-level content. Therefore, the stronger strategy is one that not only results in lower levels of repeated struggle on proximate material but also facilitates higher on-grade lesson completion.

In order to compare lesson completion among students with similar rates of repeated struggle, students in the sample were evenly split into thirds based on their per lesson rate of repeated struggle between 2020-2022. Low repeated strugglers were operationalized as those in the bottom third of repeated strugglers, medium repeated strugglers as the middle third and high repeated strugglers as the top (see Figure 5).

Figure 5. Diagram of two-step bucketing for the analysis of completion rates among similarly struggling students, by frequency of remediation, with 10% remediation threshold



Next, students were bucketed based on the percentage of all interventions that were remediation instead of learning acceleration based on the type of content they were assigned, in response to repeated struggle. This design allowed comparison of students who received similar amounts of below grade-level intervention, that differed principally by the composition of that intervention.

Because the Zearn Math platform has multiple layers of built-in learning acceleration that a teacher must bypass in order to provide students with remediation content,³ a student was categorized as “frequently remediated” if they were assigned remediation content by a teacher more than 10% of the time in response to repeated struggle. On average, students with the same rate of repeated struggle who are remediated 10% of the time, or more, complete half as many lessons in a year. Among high repeated strugglers, those who are frequently remediated complete only $\frac{1}{3}$ as many grade-level lessons in a year (see Results Table 2 for more details).

In addition to this low remediation threshold, completion rates were examined across the tiered set of remediation thresholds of <20%, 20-40% and 40%+, which illustrated the same trend, that students who spent a larger amount of their intervention time working on below grade-level material through a remediation rather than learning acceleration approach, completed fewer grade-level lessons. Similar to findings in *Accelerate, Don't Remediate*, students in this analysis completed fewer grade-level lessons when a higher proportion of the interventions they received were remediation.

Question 2: Rates and efficacy of learning acceleration for students in low-income schools and those with majority Black or Latino populations

After finding analytic support for the efficacy of learning acceleration over remediation among students broadly, the analysis focused on students in schools with majority disadvantaged populations to explore whether learning acceleration shows similar levels of efficacy in majority Black or Latino and/or low-income schools.

Zearn does not collect demographic data from individual students. Therefore in order to conduct subgroup analysis, school identifiers in the Zearn database were matched to publicly available school-level demographics, linking individual student data to school composition. Results from this analysis do not speak to the experiences of individuals with specific demographic characteristics, but rather to individuals who attend schools with majority Black, Latino and/or low-income populations.

³ On Zearn, a student who answers incorrectly receives built-in learning acceleration to support their learning. If a student continues to repeatedly struggle and needs additional support, the Zearn platform recommends to teachers foundational lessons that follow a learning acceleration approach. In order to assign below grade-level lessons that are isolated from grade-appropriate learning (“remediation”), a teacher must manually bookmark or assign students to below grade-level lessons not suggested by the foundational lesson guide.

Because time invariant characteristics drop out of fixed effects models, the sample was segmented by school-level demographic subgroups prior to the application of fixed effects logistic models for within student analysis. The sample was disaggregated into high-, mid- and low-income schools based on cutoffs of <40%, 40-75% and 75%+ economically disadvantaged, respectively.⁴ Models utilized for Question 1 were used to analyze each segmented subgroup. Once again, the aextlogit wrap was used to estimate the average change in probability of subsequent repeated struggle following remediation or learning acceleration. Figures 4 and 5 illustrate how the overall sample was disaggregated and analyzed.

Figure 6. Disaggregation and analysis of full sample into subsamples by school income level

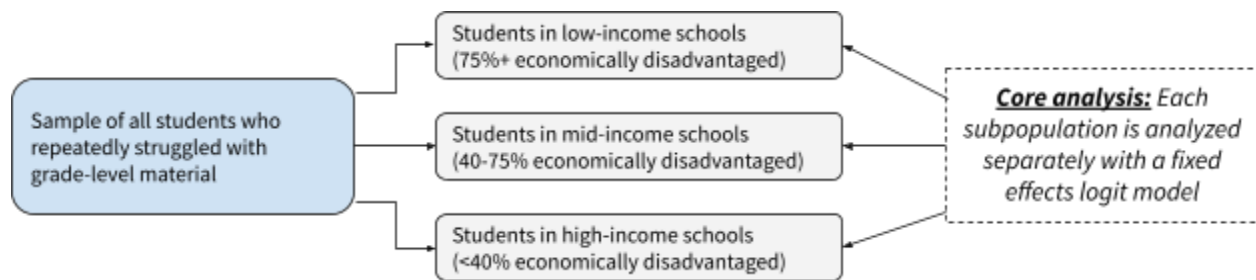
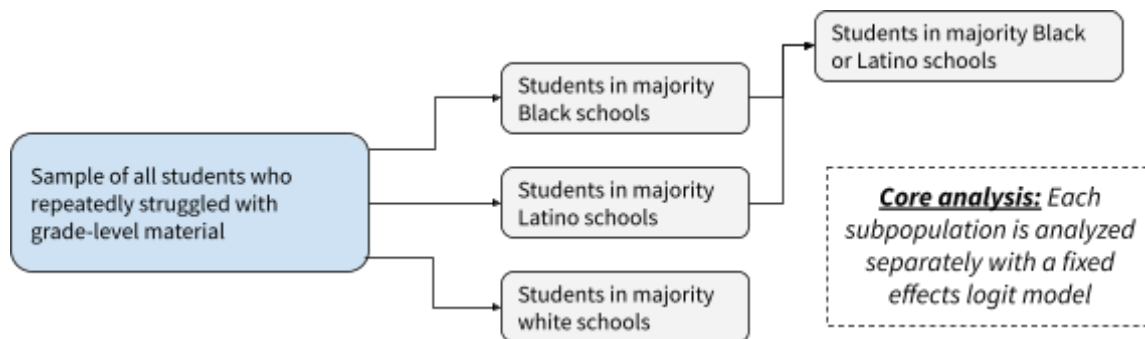


Figure 7. Disaggregation and analysis of full sample into subsamples by school-level demographics



In schools whose populations are majority low-income, Black and/or Latino, the positive impact of learning acceleration, relative to remediation, was consistent. Findings showed that among students in majority Black or Latino schools, remediation interventions to students' repeated struggle were associated with a 50% rate of repeated struggle in subsequent lessons, while learning acceleration resulted in a 40% probability of repeated struggle on the subsequent grade-level lesson.

Among students in low-income schools, remediation interventions were associated with a 51% rate of repeated struggle in subsequent lessons, while learning acceleration resulted in a 41% probability of

⁴ The cutoff of 40% was selected to differentiate between high- and mid-income schools because schools are eligible for Title I funding if their student populations are at least 40% economically disadvantaged (U.S. Department of Education, 2018). The 75%+ cutoff between mid- and low-income schools was selected because the National Center for Education Statistics defines "high poverty" schools as those in which 75% of students are free- or reduced-lunch eligible (2022).

repeated struggle. In other words, learning acceleration reduced the probability of subsequent repeated struggle by nearly 20% in both majority Black or Latino and low-income schools (see Results Table 1 for more details).

Because of the outsized efficacy of learning acceleration for the most disadvantaged students, the final analysis focused on the frequency with which students in schools with disadvantaged populations received learning acceleration instead of remediation. This analysis used the same categories of low-, medium- and high-repeated strugglers and the sample was once again segmented into subgroups based on school-level characteristics. The average proportion of interventions that were remediation (vs. learning acceleration) for low-, medium- and high-repeated strugglers was calculated within each subgroup.

Findings showed that among medium- and high-repeatedly struggling students, rates of remediation were similar across race/ethnicity and economic school demographic categories, differing by no more than 2% (see Results Table 3 for details). However, among low repeated strugglers, those who demonstrated success with grade-level content, students in majority Black, Latino and/or low-income schools were almost twice as likely to receive remediation as similar students in majority white and high-income schools. This form of over-remediation occurred at much higher rates in low-income and/or majority Black or Latino schools, despite the apparent outsized negative effects of remediation for these students.

This finding lends increased importance to the broader adaptation of learning acceleration as a strategy for all students, particularly for those in disadvantaged settings. It also raises questions about the role of bias in informing intervention decisions, particularly for students with the lowest levels of repeated struggle. These questions should be taken up in future analyses.

Limitations

This analysis produced promising evidence of the efficacy of learning acceleration over remediation when students repeatedly struggle, but limitations to the study are present. While Zearn's usage data provides an in-depth look at how students learn on the Zearn platform, it does not capture the strategies a teacher employs within the classroom outside of Zearn. Teachers likely implement a number of strategies during class time, which are not captured in the Zearn database. The impact of these in-class interventions cannot be ruled out.

Teacher behavior is informed by numerous in-person interactions with students that vastly outnumber the interactions students have with digital learning. Teachers base instructional decisions on a breadth of information that is not captured by Zearn usage data alone. Quasi-experimental fixed effects methods hold an individual constant in the analysis, but they do not preclude the possibility that teachers selected remediation interventions over learning acceleration for reasons that confound the relationship between intervention and subsequent repeated struggle.

Questions remain about whether the impact of learning acceleration vs. remediation differed, not just because of the efficacy of the strategies themselves, but because the majority of learning acceleration interventions in the data set were selected by the Zearn platform whereas all remediation interventions were selected by teachers. This difference could not be disentangled from the interventions themselves in this particular data set, but should be taken up in the future.

Additionally, though the population of Zearn users shares broad demographic features with the larger US population of students, there may be unaccounted for differences between Zearn users and non-users nationally. For example, schools that implement Zearn may be more focused on math initiatives than the average US school and therefore the relationship between remediation and learning acceleration may not be identical in non-Zearn schools. It is hard to speculate on whether the impact of learning acceleration would be amplified, reduced or the same in non-Zearn contexts.

Even within the Zearn user-base, those who have the opportunity to receive interventions are only those who repeatedly struggle on the Zearn platform and therefore repeatedly struggle at a higher rate than that of the larger population of users. They are also the students that complete multiple lessons on the platform and not those who are repeatedly struggling to such an extent that they opt-out of the math learning experience. This, however, should be similar outside of the Zearn platform where it is predominantly the repeatedly struggling students and those who show up for instruction who receive intervention, so that should not limit generalizability to real-world contexts.

Finally, while the outcome measure for this analysis is in-app repeated struggle, other outcomes of interest, such as overall proficiency as measured on state or national assessments, would need to be addressed in future inquiries using data outside of the Zearn database. While this analysis aims to support teachers in making the best decisions for their students, it is offered as a tool or framework, not as a substitute for the informed judgment of an in-classroom teacher.

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Descriptive Table 1. Total lessons completed		
	Number of lessons	Percent of lessons
On-grade lesson repeated struggle		
No repeated struggle	23,206,818	81%
Repeated struggle	5,426,878	19%
Total lessons completed	28,633,696	
Strategy		
Built-in learning acceleration	4,935,636	93%
Teacher-initiated learning acceleration	40,545	1%
Teacher-initiated remediation	302,950	6%

Descriptive Table 2. Breakdown of strategy directly following a students' final grade-level lesson, in cases where student had repeated struggle on last grade-level lesson completed		
Strategy		
Built-in learning acceleration	9,550	84%
Teacher initiated learning acceleration	49	0%
Teacher initiated remediation	1,726	15%

Descriptive Table 3. Student composition by year			
	2021	2022	Total
All students*	359,801	220,091	673,414
	*93,522 were in both 2021 & 2022		
Grade			
1	45,233	30,390	75,623
2	83,243	47,547	130,790
3	103,996	62,668	166,664
4	81,288	50,652	131,940
5	85,496	52,731	138,227
6	4,565	13,454	18,019
7	2,741	9,410	12,151
School economic composition			
High-income schools (0-40% FRL)	105,492	50,948	156,440
Mid-income schools (40-75% FRL)	139,662	83,332	222,994
Low-income schools (75%+ FRL)	115,259	85,195	200,454
Unknown	37,648	38,117	75,765
Majority population of student's school by race/ethnicity			
Students in majority AAPI schools	4,175	963	5,138
Students in majority AIAN schools	704	730	1,434
Students in majority Black schools	56,532	39,458	95,990
Students in majority Latino schools	79,750	56,132	135,882
Students in majority white schools	172,461	92,463	264,924
Unknown	84,439	67,846	152,285

Descriptive Table 4. Sample grade-level lessons with repeated struggle, by year			
	2021	2022	Total
All lessons	3,296,018	1,983,113	5,279,131
Grade			
1	326,770	190,505	517,275
2	519,786	272,491	792,277
3	890,820	480,155	1,370,975
4	730,004	421,757	1,151,761
5	778,031	465,533	1,243,564
6	34,616	91,574	126,190
7	15,991	61,098	77,089
School economic composition			
High-income schools (0-40% FRL)	742,774	334,563	1,077,337
Mid-income schools (40-75% FRL)	1,229,274	658,223	1,887,497
Low-income schools (75%+ FRL)	996,681	669,530	1,666,211
Unknown	317,986	310,693	628,679
School race/ethnicity composition			
Majority AAPI schools	27,491	4,649	32,140
Majority AIAN schools	4,974	4,345	9,319
Majority Black schools	591,673	381,326	972,999
Majority Latino schools	693,699	433,337	1,127,036
Majority white schools	696,584	539,794	1,236,378
Unknown	1,272,294	609,558	1,881,852

Descriptive Table 5. Teacher interventions for repeated on-grade struggle			
	Learning Acceleration	Remediation	Total
All lessons	4,976,181	302,950	5,279,131
Grade			
1	510,329	6,946	517,275
2	737,347	54,930	792,277
3	1,308,201	62,774	1,370,975
4	1,067,822	83,939	1,151,761
5	1,168,534	75,030	1,243,564
6	116,899	9,291	126,190
7	67,049	10,040	77,089
School economic composition			
High-income schools (0-40% FRL)	1,028,241	49,096	1,077,337
Mid-income schools (40-75% FRL)	1,788,410	99,087	1,887,497
Low-income schools (75%+ FRL)	1,563,337	102,874	1,666,211
Unknown	594,313	34,366	628,679
School race/ethnicity composition			
Majority AAPI schools	31,138	1,002	32,140
Majority AIAN schools	8,733	586	9,319
Majority Black schools	909,105	63,894	972,999
Majority Latino schools	1,065,970	61,066	1,127,036
Majority white schools	1,792,320	89,532	1,881,852
Unknown	1,167,035	69,343	1,236,378

Results Table 1. aextlogit and traditional fixed effects logistic model coefficients for probability of repeated struggle following remediation vs. learning acceleration strategies

	Mean learning acceleration intervention post-lesson rpt struggle rate using aextlogit	Remediation (dummy) using aextlogit	Remediation (dummy) traditional coefficient using xtlogit	Number of students	On-grade lessons w. repeated struggle	Learning Acceleration	Remediation
All students	0.36345186	.074859* (.0031032)	.1176015* (.0048749)	626,653	5,018,701	4,976,181	302,950
School economic composition							
Students in high-income schools (<40% FRL)	0.28504915	.0140928 (.0088716)	.0197115 (.0124087)	147,857	1,077,337	1,028,241	49,096
Students in mid-income schools (40-75% FRL)	0.35865329	.0759368* (.0052675)	.118402* (.0082129)	207,948	1,887,497	1,788,410	99,087
Students in low-income schools (75%+ FRL)	0.41045042	.097395* (.0047524)	.1652025* (.0080603)	185,156	1,666,211	1,563,337	102,874
School race/ethnicity composition							
Students in majority Black schools	0.43753488	.1241743* (.0055851)	.2207681* (.0099276)	86,476	972,999	909,105	63,894
Students in majority Latino schools	0.37910887	.0601175* (.0064592)	.097707* (.0104977)	125,757	1,127,036	1,065,970	61,066
Students in majority Black or Latino schools	0.4047726	.0961208* (.0042579)	.1626929* (.0072062)	212,187	2,100,035	1,975,075	124,960
Students in majority white schools	0.31014022	.0457665* (.0062653)	.0663418* (.009082)	251,941	1,881,852	1,792,320	89,532

Results Table 2. Lesson completion rates for similarly struggling students based on remediation as a % of all interventions

	Remediation rate	Avg Lesson completion	N's
10% remediation threshold			
Low repeated strugglers	<10%	58	166,952
	10%+	32	41,723
Med repeated strugglers	<10%	52	168,421
	10%+	31	40,881
High repeated strugglers	<10%	38	128,400
	10%+	13	80,276
Tiered thresholds			
Low repeated strugglers	<20%	59	173,759
	20-40%	52	9,123
	40%+	11	25,793
Med repeated strugglers	<20%	52	184,247
	20-40%	25	14,221
	40%+	10	10,834
High repeated strugglers	<20%	36	506,754
	20-40%	13	45,293
	40%+	4	74,606

Results Table 3. Rate of remediation for similarly struggling students across multiple school composition factors

	Low repeated strugglers	Medium repeated strugglers	High repeated strugglers
School economic composition			
Students in high-income schools (<40% FRL)	8%	6%	14%
Students in mid-income schools (40-75% FRL)	11%	7%	15%
Students in low-income schools (75%+ FRL)	15%	8%	16%
School race/ethnicity composition			
Students in majority Black schools	16%	8%	16%
Students in majority Latino schools	13%	7%	16%
Students in majority Black or Latino schools	15%	8%	16%
Students in majority white schools	9%	6%	15%